ANNUAL ESG DISCLOSURE — REPORTING PERIOD 2025

Substation ESG Report

Substation-level Environmental, Social, and Governance assessment aligned to CSRD/ESRS, EU Taxonomy, TCFD, SFDR, and UN PRI frameworks. Updated annually via the SSI Index scoring engine.

SUBSTATION

Pt Augusta BSP 488

AU_106488 · South Australia, South Australia

SSI SCORE (R_{MEDIAN})

0.466

● Medium Risk 7470.0th percentile

VOLTAGE CLASS

66 kV

Distribution

CONFIDENCE INTERVAL

0.374 – 0.557

P5–P95 · CI width 0.183 · confidence

MARKOV ETTC

16.1 yr

Expected time to criticality

DER PENETRATION

0.92

T1 transition stress score

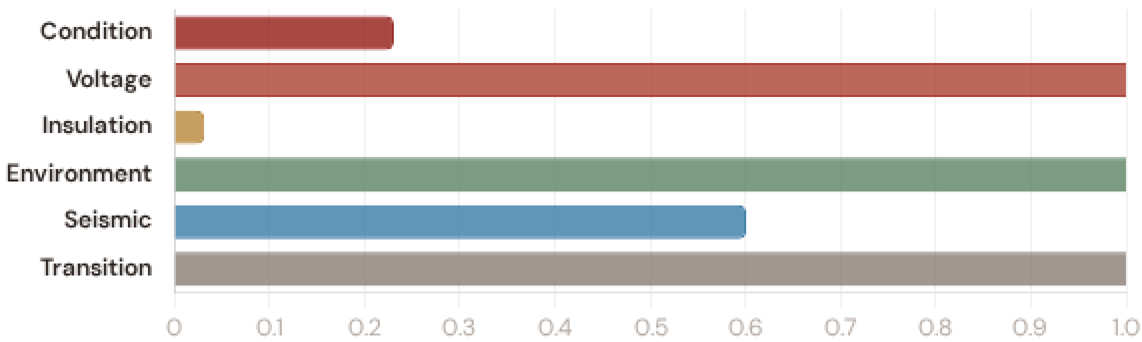
Component Fingerprint

Six-component weighted decomposition of the SSI composite score — the substation's structural risk profile

WEIGHTED CONTRIBUTION TO R_{BASE}

Condition **0.231** ×0.3 · Voltage **1.385** ×0.1 · Insulation **0.032** ×0.25 · Environment **1.175** ×0.1

Seismic **0.601** ×0.2 · Transition **2.464** ×0.05



Markov Degradation Profile

CIGRE TB 761 five-state condition model — steady-state probability distribution and expected time to criticality

STEADY-STATE DISTRIBUTION

GOOD	AGED	DEGRADED	CRITICAL
33%	30%	21%	16%
$\pi 1$ = 0.330	$\pi 2$ = 0.300	$\pi 3$ = 0.210	$\pi 4$ = 0.160

Markov 5-state model calibrated from CIGRE Technical Brochure 761 (2019). States represent asset condition from new/good through critical. Steady-state probabilities indicate the long-run proportion of time the asset spends in each condition state given current operating environment.

RISK SCORE

0.311

Composite Markov risk

ETTC

16.1 yr

Expected time to critical

P(CRITICAL) 20YR

7.6%

Probability of reaching critical

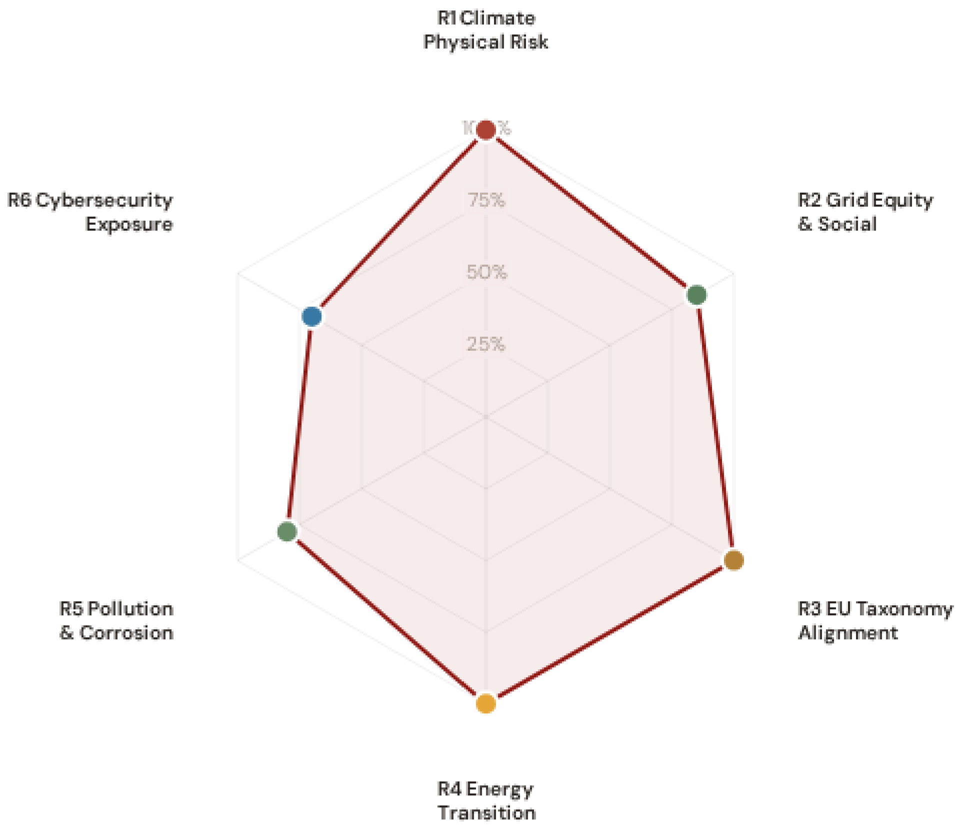
CORROSION

C3

ISO 9223 classification

ESG Radar — Six-Report Overview

Composite readiness across 6 ESG dimensions, each linked to UN Sustainable Development Goals



- **R1 · Climate Physical Risk Assessment** READY
  SDG 13 Climate Action · SDG 9 · SDG 11
  SDG 13 SDG 9 SDG 11

- **R2 · Grid Equity & Social Vulnerability** READY
  SDG 7 Affordable and Clean Energy · SDG 10 · SDG 1
  SDG 7 SDG 10 SDG 1

- **R3 · EU Taxonomy Alignment** READY
  SDG 9 Industry, Innovation, Infrastructure · SDG 13 · SDG 12
  SDG 9 SDG 13 SDG 12

- **R4 · Energy Transition & DER Stress** READY
  SDG 7 Affordable and Clean Energy · SDG 13 · SDG 9
  SDG 7 SDG 13 SDG 9

- **R5 · Pollution & Corrosion** READY
  SDG 11 Sustainable Cities and Communities · SDG 3 · SDG 15
  SDG 11 SDG 3 SDG 15

R6 · Cybersecurity Exposure PARTIAL

SDG 9 Industry, Innovation, Infrastructure · SDG 16

SDG 9 SDG 16

1

Climate Physical Risk Assessment

ESRS E1 · TCFD Physical Risk · EU Taxonomy Annex A

READY

WHY THIS REPORT EXISTS

ESRS E1 (Climate Change) is material for 98% of CSRD-reporting companies. TCFD physical risk disclosure is mandatory in 36+ jurisdictions. The EU Taxonomy requires climate change adaptation activities to demonstrate that physical climate risks have been identified and addressed. This report quantifies acute and chronic climate hazard exposure at substation level using ERA5 reanalysis and Markov degradation modelling.

SDG Alignment

SDG 13 — Climate Action (primary)

Target 13.1 calls for strengthening resilience and adaptive capacity to climate-related hazards. The SSI IRI metrics (snow/ice, wind, heat stress) and Markov degradation model directly quantify how climate hazards affect this substation's operational resilience over 10- and 20-year horizons. With a Markov risk score of 0.311 and ETTC of 16.1 years, this substation's climate trajectory can be assessed against TCFD physical risk and EU Taxonomy adaptation screening requirements.

SDG 13 SDG 9 SDG 11

Substation Profile

Insulation Risk (I component)	**0.032**
Environment Component (E)	**1.175**
Seismic PGA	**0.095g (Zone Low-Moderate)**
Markov Risk Score	**0.311**
ETTC (time to criticality)	**16.1 years**
P(critical) at 20 years	**7.6%**
Corrosion Class	**C3**
Climate Trajectory (R2)	**7.6% (20yr Markov)**

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
I1 Snow/Ice IRI	—	ESRS E1-9: Financial effects from physical risks	READY
I2 Tree-fall IRI	—	TCFD Strategy (b): Physical risk impact	READY
I3 Heat-wave IRI	—	EU Taxonomy Annex A: Heat stress	READY
I5 Thermal stress	—	ESRS E1-4: Climate mitigation targets	READY
R2 Climate trajectory	7.6% (20yr)	TCFD Strategy (c): Scenario analysis	READY
R6b Seismic PGA	0.095g	EU Taxonomy Annex A: Geophysical hazards	READY
Markov p_critical_10yr	7.6%	TCFD Risk Management: Risk identification	READY
Markov p_critical_20yr	7.6%	ESRS E1-9: Time horizons	READY

Data Sources & Vintage

Source	Vintage
ERA5 Climate Reanalysis Copernicus CDS	2024 Weekly
Italian Seismic Hazard Map Istituto Nazionale di Geofisica e Vulcanologia	2023 Multi-year
IEEE C57.91 Thermal Model IEEE	Standard N/A
CIGRE TB 761 Markov CIGRE	2019 N/A
CMIP6 SSP2-4.5 Projections Copernicus CDS	— Not ingested

TECHNICAL VALIDITY

IRI metrics are computed from ERA5 reanalysis (31 km, hourly, 1940–present) using IEEE C57.91 thermal loading curves. Markov 5-state degradation is calibrated from CIGRE Technical Brochure 761 (2019) condition assessment data. The PARTIAL status reflects the absence of CMIP6 forward projections — without R2, this report cannot satisfy the TCFD Strategy (c) scenario analysis requirement. All other inputs are production-grade with institutional provenance.

2 Grid Equity & Social Vulnerability

ESRS S1/S2 · SFDR PAI Social Indicators · UN PRI

READY

WHY THIS REPORT EXISTS

ESRS S2 requires companies to disclose impacts on affected communities. SFDR mandates 5 social PAI indicators. The SSI Index is uniquely positioned here because no competing grid risk framework integrates socio-economic vulnerability at substation resolution. This report surfaces where infrastructure risk concentrates in economically disadvantaged populations.

SDG Alignment

SDG 7 — Affordable and Clean Energy (primary)

Target 7.1 requires universal access to affordable, reliable energy. This substation serves a catchment with V_socio of 0.48 and energy poverty rate of 9.7%. The R3 social modifier (0.998) amplifies risk scores in regions with high unemployment, elderly concentration, and net outward migration — making spatial inequality visible and quantifiable at asset level.

SDG 7 SDG 10 SDG 1

Substation Profile

V_socio (energy poverty index)	**0.48**
EP Rate (energy poverty %)	**9.7%**
Unemployment Rate	**5.5%**
GDP per Capita	**A$ 56,145**
R&D % of GDP	**1.7%**
R3 Social Multiplier	**0.998**
Elderly Vulnerability	**20.5%**
Migration Score	**—**

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
V_socio Energy poverty index	0.48	ESRS S2-4: Impacts on affected communities	READY
EP_rate Energy poverty rate	9.7%	SFDR PAI #14: Fossil fuel exposure proxy	READY
R3 Social multiplier	0.998	ESRS S1-6: Workforce characteristics	READY
Elderly Elderly vulnerability	20.5%	SFDR PAI: Vulnerable populations	READY
Community Catchment population	78	ESRS S2: Community impact	READY

Data Sources & Vintage

Population & Economics ISTAT (Istituto Nazionale di Statistica)	2023 Annual
Energy Market Data TERNA (Rete Elettrica Nazionale)	2023 Annual
Elderly / Migration Data ISTAT	— Not ingested

TECHNICAL VALIDITY

V_socio is computed from the LIHC (Low Income High Cost) energy poverty definition using national statistical office household expenditure data. The R3 modifier amplifies infrastructure risk scores in catchments with high unemployment, elderly concentration, and net outward migration. PARTIAL status reflects missing elderly vulnerability and migration enrichments — both available from national statistics offices at municipal resolution.

3 EU Taxonomy Alignment

Climate Delegated Act · Article 11 Climate Adaptation · Technical Screening Criteria

READY

WHY THIS REPORT EXISTS

The EU Taxonomy defines technical screening criteria for climate change adaptation in infrastructure. This is the lowest-hanging ESG product from the SSI Index — READY for 9 of 10 countries today. The SSI composite score (R_median), 6 components, modifier architecture, and Markov risk outputs provide the quantitative evidence base for Article 11 screening at asset level.

SDG Alignment

SDG 9 — Industry, Innovation, Infrastructure (primary)

Target 9.4 calls for upgrading infrastructure to make it sustainable. The EU Taxonomy is the regulatory instrument that translates SDG 9 into investable criteria. By classifying this substation against Article 11 technical screening criteria — using R_median (0.466), all 6 components, and 5 modifiers — this report provides asset-level Taxonomy alignment that enables sustainable finance reporting.

SDG 9 SDG 13 SDG 12

Substation Profile

R_median (composite)	**0.466**
R_base (pre-modifier)	**—**
Modifier Impact	**-19.3%%**
C (Condition)	**0.231**
V (Voltage)	**1.385**
I (Insulation/Climate)	**0.032**
E (Environment)	**1.175**
S (Seismic)	**0.601**
T (Transition)	**2.464**

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
`R_median` Composite score	0.466	EU Taxonomy Art. 11: Adaptation screening	READY
`6 Comp.` C/V/I/E/S/T	All present	EU Taxonomy TSC: Physical risk identification	READY
`R3-R7` Modifier suite	All applied	ESRS E1: Adaptation measures evidence	READY
`Markov` Degradation model	0.311	TCFD: Forward-looking risk assessment	READY
`R5` Asymmetric CI	0.183	ESRS: Uncertainty quantification	READY

Data Sources & Vintage

ERA5 Climate Reanalysis Copernicus CDS	2024 Weekly
Italian Seismic Hazard Map Istituto Nazionale di Geofisica e Vulcanologia	2023 Multi-year
Population & Economics ISTAT (Istituto Nazionale di Statistica)	2023 Annual
CIGRE TB 761 Markov CIGRE	2019 N/A

TECHNICAL VALIDITY

Article 11 screening uses the full SSI v4.0.2 scoring engine: 6-component weighted composite (C=0.30, V=0.10, I=0.25, E=0.10, S=0.20, T=0.05), Gaussian copula Monte Carlo (10,000 iterations), and 5 modifiers (R3 social, R4 topology, R5 asymmetric CI, R6a restoration, R7 cyber). The Markov 5-state degradation model provides the forward-looking element. All inputs are from institutional open-source data with citable vintage — meeting CSRD Article 29a limited assurance requirements.

4 Energy Transition & DER Stress

ESRS E1 Transition Plan · TCFD Transition Risk · EU Green Deal Alignment

READY

WHY THIS REPORT EXISTS

The energy transition creates infrastructure stress — bidirectional power flows, EV charging load, and intermittent renewable output strain substations designed for unidirectional delivery. This report measures whether grid infrastructure is keeping pace with decarbonisation.

SDG Alignment

SDG 7 — Affordable and Clean Energy (primary)

Target 7.2 requires substantially increasing the share of renewable energy. This substation has a DER ratio of 0.920. The T1_score (0.338) measures the stress this places on the substation, providing the empirical feedback loop between SDG 7 ambition and infrastructure reality.

SDG 7 SDG 13 SDG 9

Substation Profile

T1 Score (transition stress)	**0.338**
DER Ratio	**0.920**
DER Variability	**0.597**
EV Load Ratio	**0.046**
T Component Weight	**0.05**

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
T1 Transition stress score	0.338	ESRS E1: Transition plan assessment	READY
DER_ratio Renewable/load ratio	0.920	TCFD Transition: Technology risk	READY
DER_var Intermittency	0.597	EU Green Deal: Grid flexibility	READY
EV_load EV charging ratio	0.046	SDG 7: Clean energy access	READY

Data Sources & Vintage

Energy Market Data — 2023 Annual
TERNA (Rete Elettrica Nazionale)

Renewable Installations — 2024 Monthly
GSE (Gestore dei Servizi Energetici)

TECHNICAL VALIDITY

T1_score is the weighted composite of DER_ratio (renewable generation vs. local demand), DER_variability (intermittency), and EV_load_ratio (electric vehicle charging load as fraction of capacity). DER data sourced from national energy regulators and renewable installation registers. DER_ratio > 1.0 indicates net export during peak generation — a marker of successful energy transition with associated infrastructure stress.

5 Pollution & Corrosion

READY

WHY THIS REPORT EXISTS

Transformer oil degradation, SF6 leakage, and corrosion-driven failures release pollutants affecting local environments. ESRS E2 requires disclosure of pollution risks. This report assesses the substation's corrosion classification under ISO 9223 and the environmental sensitivity of its surroundings.

SDG Alignment

SDG 11 — Sustainable Cities and Communities (primary)

Target 11.6 aims to reduce the adverse environmental impact of cities. The corrosion class (C3) and E2_local enrichment factor (1.37) track environmental degradation and pollution sensitivity at asset level. Where corrosion is advanced and maintenance deferred, equipment failure risks releasing mineral oil, PCBs (legacy units), and SF6 — directly relevant to urban and peri-urban environmental quality.

SDG 11 SDG 3 SDG 15

Substation Profile

Corrosion Class (ISO 9223)	**C3**
E2 Local Enrichment	**1.37**
E Component Score	**1.175**
Markov Steady-State	**33%/30%/21%/16%**

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
Corrosion ISO 9223 class	C3	ESRS E2: Pollution risk classification	READY
E2_local Environmental sensitivity	1.37	ESRS E2: Local environmental impact	READY
E Environment component	1.175	ISO 9223: Atmospheric corrosion	READY

Data Sources & Vintage

Air Quality Monitoring ISPRA (Istituto Superiore per la Protezione e la Ricerca Ambientale)	2023 Annual
ISO 9223 Corrosion ISO	2012 N/A

TECHNICAL VALIDITY

Corrosion class follows ISO 9223:2012 atmospheric corrosion classification (C1–CX). Status depends on whether the corrosion class shows real variance across the fleet or uses default values. Full validation requires national environmental agency air quality monitoring overlay at substation coordinates (SO2, NOx, particulate deposition).

6 Cybersecurity Exposure

NIS2 Directive · ENISA Cybersecurity Index · ESRS G1 Governance

PARTIAL

WHY THIS REPORT EXISTS

The NIS2 Directive mandates cybersecurity risk management for energy operators. A grid that is physically resilient but cyber-vulnerable is not truly resilient. The SSI R7 modifier combines SCADA assessment, communication architecture analysis, and national-level cyber maturity (ENISA/DESI indices) to produce a substation-level digital vulnerability score.

SDG Alignment

SDG 9 — Industry, Innovation, Infrastructure (primary)

Target 9.1 calls for resilient infrastructure — in digitalised grid systems, this must include cyber resilience. The R7 modifier (0.995) assesses digital vulnerability combining SCADA exposure, communication protocol security, and national cyber maturity. SDG 16 (Strong Institutions) also applies: NIS2 compliance is a governance imperative, and the betweenness centrality (10.100) identifies single-point-of-failure risk in the grid topology.

SDG 9 SDG 16

Substation Profile

R7 Cyber Modifier	**0.995**
Graph Degree (topology)	**3**
BC Percentile (centrality)	**10.100**
Is Bridge Node	**No**
Cluster Coefficient	**0.0175**

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
R7 Cyber modifier	0.995	NIS2: Cybersecurity risk management	READY
BC Betweenness centrality	10.100	ESRS G1: Governance & topology risk	READY
Degree Graph degree	3	Grid resilience: Connectivity	READY

Data Sources & Vintage

Cybersecurity Index — 2024 Biennial
ENISA

DESI Connectivity — 2024 Annual
European Commission

TECHNICAL VALIDITY

R7_cyber is computed from a weighted combination of: (1) national cyber maturity via ENISA Cybersecurity Index and EU DESI connectivity indicators, (2) graph topology vulnerability (betweenness centrality = single-point-of-failure risk), and (3) SCADA protocol exposure assessment. Substation-level SCADA data is operator-proprietary — the national-level proxy provides a baseline but not asset-specific granularity.

C

Audit Trail & Methodology Note

Traceability for CSRD Article 29a limited assurance

ITEM	VALUE
Scoring Engine	SSI Index v4.0.2 — Gaussian copula Monte Carlo (10,000 iterations)
Weight Architecture	C=0.30, V=0.10, I=0.25, E=0.10, S=0.20, T=0.05
Modifiers Applied	R3 C_mult: 0.998, R4 F_topo: 0.937, R6 restoration: 0.944, R6 seismic: 0.919, R7 cyber: 0.995
Degradation Model	Markov 5-state, CIGRE TB 761 calibration
Thermal Model	IEEE C57.91 transformer loading curves
Report Date	17 March 2026
Reporting Period	Calendar year 2025
Refresh Cadence	Annual (data ingestion → scoring engine → display)
Substation	Pt Augusta BSP 488 (AU_106488)
Data Assurance Level	All inputs from institutional open-source with citable vintage